BCS-086	Advanced Computer Architecture	L T P 3 0 0	3 Credits

Pre-requisites: None

Course Objectives:

CO1	To make students know about the Parallelism concepts
CO2	To give the students an elaborate idea about the high-performance memory systems
CO3	To introduce the advanced processor architectures to the students
CO4	To make the students know about the importance of multiprocessor and multi computers and different programming models
CO5	To understand Memory hierarchy technology

Detailed Syllabus

MODULE-I

Introduction and performance: Evolution of computer Architecture, Architectural classification schemes and parallel computing models, conditions of parallelism, program flow mechanisms, performance evaluation and speedup performance laws, RISC and CISC processors, VLIW architecture

MODULE-II

Pipelining: Instruction level parallelism, principles of linear and nonlinear pipelining Techniques, Hazards, Instruction and arithmetic pipeline design, super scalar and super pipeline design.
Memory hierarchy technology: Cache memory organizations and performance issues; multilevel caches, Virtual memory technology and memory management.

MODULE-III

SIMD processor: SIMD array processor, Interconnection networks, SIMD matrix multiplication algorithm, vector processor architecture and instruction types.
MIMD multiprocessor: shared and distributed memory architectures, cache coherence and Synchronization.

Course Outcomes: After the completion of the course the student will be able to:

CO1	Demonstrate concepts of parallelism in hardware/software.
CO2	Discuss memory organization and mapping techniques
CO3	Describe architectural features of advanced processors.
CO4	Interpret performance of different pipelined processors.
CO5	Discuss Memory hierarchy technology

Text Books:

1.Kai Hwang, “Advanced Computer Architecture,” McGraw-Hill.
2. Hwang and Briggs, “Computer Architecture and Parallel Processing,” McGraw Hill.

Reference Books:

1. Pipelined and Parallel processor design by Michael J. Fiynn – 1995, Narosa.
2. Computer Architecture A quantitative approach 3rd edition John L. Hennessy & David A. Patterson Morgan Kufmann (An Imprint of Elsevier)